

矩阵

线性方程组

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \\ \dots \dots \dots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_n \end{cases}$$

的系数 a_{ij} ($i, j = 1, 2, \dots, n$) 和右端项 b_j ($j = 1, 2, \dots, n$) 按原位置构成一数表:

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_n \end{bmatrix}$$

定义. 由 $m \times n$ 个数 a_{ij} ($i = 1, 2, \dots, m; j = 1, 2, \dots, n$) 排成的 m 行、 n 列数表称为 m 行 n 列矩阵, 简称 $m \times n$ 矩阵

记作

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} = A_{m \times n} = (a_{ij})_{m \times n}$$

元素为实数的矩阵称为实矩阵, 元素为复数的矩阵称为复矩阵, 所有元素均为 0 的矩阵称为零矩阵, 记作 O

- 若 A 的行数与列数都等于 n , 则称 A 为 n 阶方阵, 记为 A_n
(行列式里的数组成的矩阵就是方阵)
- 如果两个矩阵具有相同的行数和列数, 则称这两个矩阵为同型矩阵
- 若 A 与 B 为同型矩阵且对应元素相等, 则 $A = B$

特殊矩阵:

(1) 仅一行 (行矩阵 / 行向量)

$$A = [a_1 \ a_2 \ \dots \ a_n]$$

(2) 仅一列 (列矩阵 / 列向量)

$$B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

(3) n 阶单位矩阵

$$E = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}$$

(4) 对角矩阵

$$\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$$

$$= \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix}$$

上三角矩阵

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ & a_{22} & \dots & a_{2n} \\ & & \ddots & \vdots \\ & & & a_{nn} \end{bmatrix}$$

下三角矩阵

$$\begin{bmatrix} a_{11} & & & \\ a_{21} & a_{22} & & \\ \vdots & & \ddots & \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}$$

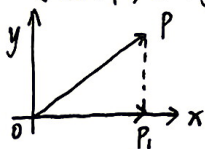
线性变换的概念

$$\begin{cases} y_1 = a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \\ y_2 = a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n \\ \dots \\ y_m = a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n \end{cases} \quad \text{称为从变量 } x_1, x_2, \dots, x_n \text{ 到变量 } y_1, y_2, \dots, y_n \text{ 的线性变换}$$

$$A = (a_{ij})_{m \times n} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \quad \text{为线性变换的系数矩阵}$$

例如, $\begin{cases} y_1 = x_1 \\ y_2 = x_2 \\ \dots \\ y_n = x_n \end{cases}$ 是恒等变换, 其系数矩阵是单位矩阵

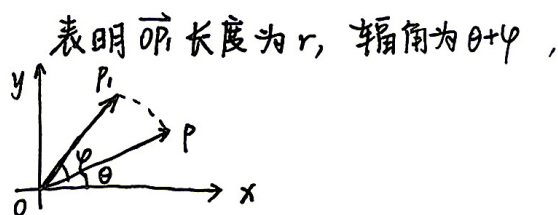
▶ 矩阵 $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ 所对应的线性变换 $\begin{cases} x_1 = x \\ y_1 = 0 \end{cases}$ 可看作是 xoy 面上将向量 $\vec{OP} = \begin{bmatrix} x \\ y \end{bmatrix}$ 变为 $\vec{OP}_1 = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} x \\ 0 \end{bmatrix}$ 的变换, 这是一个投影变换



▶ 矩阵 $\begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix}$ 所对应的线性变换 $\begin{cases} x_1 = x\cos\varphi - y\sin\varphi \\ y_1 = x\sin\varphi + y\cos\varphi \end{cases}$ 可看作是将 xoy 面上的向量 $\vec{OP} = \begin{bmatrix} x \\ y \end{bmatrix}$ 变为 $\vec{OP}_1 = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$, 设 $\vec{OP}$ 长为 r , 辐角为 θ . 即设 $x = r\cos\theta, y = r\sin\theta$

$$\begin{cases} x_1 = r(\cos\varphi\cos\theta - \sin\varphi\sin\theta) = r\cos(\theta+\varphi) \\ y_1 = r(\sin\varphi\cos\theta + \cos\varphi\sin\theta) = r\sin(\theta+\varphi) \end{cases}$$

这是一个旋转变换 ($\vec{OP}$ 逆时针旋转 φ 角)



练习.

1. 二人零和对策问题, 两人玩“石头-剪子-布”, 若规定胜者得1分, 输者得-1分, 平手各得0分. 则对各种可能的局势 (每一局势得分之和为零, 即零和), 试用矩阵表示他们的输赢状况.

解:

$$\begin{matrix} & \begin{matrix} \text{A 策略} \downarrow \\ \text{石头} \\ \text{剪子} \\ \text{布} \end{matrix} & \begin{matrix} \text{B 策略} \rightarrow \\ \text{石头} \\ \text{剪子} \\ \text{布} \end{matrix} \\ \begin{bmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix} \end{matrix}$$

2. 有6名选手参加乒乓球比赛, 成绩如下: 选手1胜2, 4, 5, 6而负于3; 选手2胜4, 5, 6而负于1, 3; 选手3胜1, 2, 4而负于5, 6; 选手4胜5, 6而负于1, 2, 3; 选手5胜3, 6而负于1, 2, 4. 若胜1场得1分, 负1场得0分, 试用矩阵表示输赢状况并排序.

$$\begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 \end{matrix} \\ \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

排序 (胜多负少) = 1 2 3 4 5 6

矩阵的加法

$$A+B = (a_{ij} + b_{ij})_{m \times n} = \begin{bmatrix} a_{11}+b_{11} & a_{12}+b_{12} & \dots & a_{1n}+b_{1n} \\ a_{21}+b_{21} & a_{22}+b_{22} & \dots & a_{2n}+b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}+b_{m1} & a_{m2}+b_{m2} & \dots & a_{mn}+b_{mn} \end{bmatrix}$$

只有两个同型矩阵才能进行加法运算

$$-A = (-a_{ij}), -A \text{ 是 } A \text{ 的负矩阵}, A+(-A)=O$$

运算律:

$$A+B = B+A$$

$$(A+B)+C = A+(B+C)$$

矩阵的减法

$$A-B = A+(-B)$$

数与矩阵的乘法 (简称数乘运算)

$$\lambda A = A\lambda = (\lambda a_{ij})_{m \times n} = \begin{bmatrix} \lambda a_{11} & \lambda a_{12} & \dots & \lambda a_{1n} \\ \lambda a_{21} & \lambda a_{22} & \dots & \lambda a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda a_{m1} & \lambda a_{m2} & \dots & \lambda a_{mn} \end{bmatrix}$$

运算律:

$$(\lambda\mu)A = \lambda(\mu A)$$

$$(\lambda+\mu)A = \lambda A + \mu A$$

$$\lambda(A+B) = \lambda A + \lambda B$$

例. 已知 $A = \begin{bmatrix} -1 & 2 & 3 & 1 \\ 0 & 3 & -2 & 1 \\ 4 & 0 & 3 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 3 & 2 & -1 \\ 5 & -3 & 0 & 1 \\ 1 & 2 & -5 & 0 \end{bmatrix}$, 求 $3A-2B$.

解: $3A-2B = 3 \begin{bmatrix} -1 & 2 & 3 & 1 \\ 0 & 3 & -2 & 1 \\ 4 & 0 & 3 & 2 \end{bmatrix} - 2 \begin{bmatrix} 4 & 3 & 2 & -1 \\ 5 & -3 & 0 & 1 \\ 1 & 2 & -5 & 0 \end{bmatrix} = \begin{bmatrix} -3-8 & 6-6 & 9-4 & 3+2 \\ 0-10 & 9+6 & -6-0 & 3-2 \\ 12-2 & 0-4 & 9+10 & 6-0 \end{bmatrix} = \begin{bmatrix} -11 & 0 & 5 & 5 \\ -10 & 15 & -6 & 1 \\ 10 & -4 & 19 & 6 \end{bmatrix}$

例. 已知 $A = \begin{bmatrix} 3 & -1 & 2 & 0 \\ 1 & 5 & 7 & 9 \\ 2 & 4 & 6 & 8 \end{bmatrix}$, $B = \begin{bmatrix} 7 & 5 & -2 & 4 \\ 5 & 1 & 9 & 7 \\ 3 & 2 & -1 & 6 \end{bmatrix}$, 且 $A+2X=B$, 求 X

解: $X = \frac{1}{2}(B-A) = \frac{1}{2} \begin{bmatrix} 7-3 & 5+1 & -2-2 & 4-0 \\ 5-1 & 1-5 & 9-7 & 7-9 \\ 3-2 & 2-4 & -1-6 & 6-8 \end{bmatrix} = \begin{bmatrix} 2 & 3 & -2 & 2 \\ 2 & -2 & 1 & -1 \\ \frac{1}{2} & -1 & -\frac{7}{2} & -1 \end{bmatrix}$

矩阵与矩阵的乘法

设 $A=(a_{ij})$ 是一个 $m \times s$ 矩阵, $B=(b_{ij})$ 是一个 $s \times n$ 矩阵, 则 $A \times B$ 是一个 $m \times n$ 矩阵 $C=(c_{ij})$, 其中 $c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{is}b_{sj} = \sum_{k=1}^s a_{ik}b_{kj}$ ($i=1, 2, \dots, m; j=1, 2, \dots, n$) 记作 $C=AB$

只有 A 的列数 = B 的行数, A 与 B 才能相乘

若 $A=(a_{ij})_{1 \times n}$ (行向量), $B=(b_{ij})_{n \times 1}$ (列向量), 则 $AB=(c_{ij})_{1 \times 1}$ 是一阶方阵, 即实数, 而 $BA=(c_{ij})_{n \times n}$ 是 n 阶方阵

例. 已知 $A=(1 \ 2 \ 3)$, $B = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$

$$AB = (1 \ 2 \ 3) \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} = 10$$

$$BA = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} (1 \ 2 \ 3) = \begin{bmatrix} 3 & 6 & 9 \\ 2 & 4 & 6 \\ 1 & 2 & 3 \end{bmatrix}$$

已知 $A = \begin{bmatrix} 2 & 1 \\ -4 & -2 \end{bmatrix}$, $B = \begin{bmatrix} 3 & -1 \\ -6 & 2 \end{bmatrix}$

则 $AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, $BA = \begin{bmatrix} 10 & 5 \\ -20 & -10 \end{bmatrix}$, $A^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

一般地, $AB \neq BA$

若 $AB = BA$, 则称 A 与 B 是可交换的

$AB = 0 \not\Rightarrow A = 0$ 或 $B = 0$

$A^2 = 0 \not\Rightarrow A = 0$

$AB = A^2 \not\Rightarrow A = B$

$EA = AE = A$

$A^1 = A$, $A^2 = A \cdot A$, ... $A^{k+1} = A^k \cdot A$

仅当 A 与 B 可交换时, 才有 $(AB)^k = A^k B^k$, $(A+B)^2 = A^2 + 2AB + B^2$, $(A+B) \cdot (A-B) = A^2 - B^2$ 等

例. 求与矩阵 $A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ 可交换的所有矩阵

解:

设与 A 可交换的矩阵是 $B = \begin{bmatrix} a & b & c & d \\ a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \end{bmatrix}$

则 $AB = \begin{bmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, $BA = \begin{bmatrix} 0 & a & b & c \\ 0 & a_1 & b_1 & c_1 \\ 0 & a_2 & b_2 & c_2 \\ 0 & a_3 & b_3 & c_3 \end{bmatrix}$

令 $AB = BA$, 则

$a_1 = 0$ $b_1 = a$ $c_1 = b$ $d_1 = c$
 $a_2 = 0$ $b_2 = a_1 = 0$ $c_2 = b_1 = a$ $d_2 = c_1 = b$
 $a_3 = 0$ $b_3 = a_2 = 0$ $c_3 = b_2 = 0$ $d_3 = c_2 = a$

$\Rightarrow B = \begin{bmatrix} a & b & c & d \\ 0 & a & b & c \\ 0 & 0 & a & b \\ 0 & 0 & 0 & a \end{bmatrix}$

(a, b, c, d 是任意实数)

运算律

$(AB)C = A(BC)$

$\lambda(AB) = (\lambda A)B = A(\lambda B)$

$A(B+C) = AB+AC$

$(A+B)C = AC+BC$

矩阵的转置

$$\text{若 } A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \text{ 则 } A^T = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{bmatrix}$$

运算律

$$(A^T)^T = A \quad (A \pm B)^T = A^T \pm B^T \quad (\lambda A)^T = \lambda A^T \quad (AB)^T = B^T A^T$$

对于方阵 A , 若 $A^T = A$, 即 $a_{ji} = a_{ij}$, 则称 A 为对称矩阵

若 $A^T = -A$, 即 $a_{ji} = -a_{ij}$, 则称 A 为反对称矩阵

设 A 与 B 是两个 n 阶反对称矩阵, 证明: AB 是反对称矩阵当且仅当 $AB = -BA$

证明: $\because A^T = -A, B^T = -B$

若 $AB = -BA$, 则 $(AB)^T = B^T A^T = (-B)(-A) = BA = -AB \quad \therefore AB$ 是反对称矩阵

若 AB 是反对称矩阵, 则 $(AB)^T = -AB$, 有 $AB = -(AB)^T = -B^T A^T = -(-B)(-A) = -BA$

设列矩阵 $X = (x_1, x_2, \dots, x_n)^T$ 满足 $X^T X = I$, E 为 n 阶单位矩阵, $H = E - 2XX^T$, 证明:

H 是对称矩阵且 $HH^T = E$

证明: $\because H^T = (E - 2XX^T)^T = E^T - 2(XX^T)^T = E - 2(X^T)^T X^T = E - 2XX^T = H$

$\therefore H$ 是对称矩阵

$$\begin{aligned} HH^T &= H^2 = (E - 2XX^T)^2 = E - 4XX^T + 4(XX^T)(XX^T) \\ &= E - 4XX^T + 4X \underbrace{(X^T X)}_I X^T = E - 4XX^T + 4XX^T = E \end{aligned}$$

方阵的行列式

由 n 阶方阵 A 的元素所构成的行列式(各元素的位置不变), 记作 $|A|$ 或 $\det A$

n 阶方阵是一个数表, 而 n 阶行列式是一个数

运算律

$$|A^T| = |A| \quad |\lambda A| = \lambda^n |A| \quad |AB| = |BA| = |A||B|$$

例. 已知 A 为三阶方阵且 $|A| = -2$, 求 $|A^2|$, $|-2A|$

解: $|A^2| = |A|^2 = 4$

$$|-2A| = (-2)^3 |A| = 16$$

1. 计算 $(x_1 \ x_2 \ x_3) \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

解: 原式 = $(x_1 a_{11} + x_2 a_{12} + x_3 a_{13} \quad x_1 a_{21} + x_2 a_{22} + x_3 a_{23} \quad x_1 a_{31} + x_2 a_{32} + x_3 a_{33}) \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$
 $= (x_1 a_{11} + x_2 a_{12} + x_3 a_{13}) x_1 + (x_1 a_{21} + x_2 a_{22} + x_3 a_{23}) x_2 + (x_1 a_{31} + x_2 a_{32} + x_3 a_{33}) x_3$
 $= a_{11} x_1^2 + a_{22} x_2^2 + a_{33} x_3^2 + 2x_1 x_2 a_{12} + 2x_1 x_3 a_{13} + x_2 x_3 a_{32} + x_2 x_3 a_{23}$

2. 已知两个线性变换

$$\begin{cases} x_1 = 2y_1 & + y_3 \\ x_2 = -2y_1 + 3y_2 + 2y_3 \\ x_3 = 4y_1 + y_2 + 5y_3 \end{cases} \quad \text{和} \quad \begin{cases} y_1 = -3z_1 + z_2 \\ y_2 = 2z_1 & + z_3 \\ y_3 = & -z_2 + 3z_3 \end{cases}, \text{求从 } z_1, z_2, z_3 \text{ 到 } x_1, x_2, x_3 \text{ 的线性变换}$$

解: $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 1 \\ -2 & 3 & 2 \\ 4 & 1 & 5 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 1 \\ -2 & 3 & 2 \\ 4 & 1 & 5 \end{bmatrix} \begin{bmatrix} -3 & 1 & 0 \\ 2 & 0 & 1 \\ 0 & -1 & 3 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} -6 & 1 & 3 \\ 12 & -4 & 9 \\ -10 & -1 & 16 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$

$$\begin{cases} x_1 = -6z_1 + z_2 + 3z_3 \\ x_2 = 12z_1 - 4z_2 + 9z_3 \\ x_3 = -10z_1 - z_2 + 16z_3 \end{cases}$$

3. 某企业某年出口到三个国家的两种货物的情况如下:

产量 货物 \ 国家	美国	德国	日本	单位价格	单位重量	单位体积
A ₁	3000	1500	2000	0.5	0.04	0.2
A ₂	1400	1300	800	0.4	0.06	0.4

解:

$$\begin{bmatrix} 3000 & 1400 \\ 1500 & 1300 \\ 2000 & 800 \end{bmatrix} \begin{bmatrix} 0.5 & 0.04 & 0.2 \\ 0.4 & 0.06 & 0.4 \end{bmatrix} = \begin{bmatrix} 2060 & 204 & 1160 \\ 1270 & 138 & 720 \\ 1320 & 128 & 720 \end{bmatrix}$$

总价值: $2060 + 1270 + 1320 = 4650$ (万元)

总重量: $204 + 138 + 128 = 470$ (吨)

总体积: $1160 + 720 + 720 = 2600$ (米³)

4. 设矩阵 $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, 求所有与A可交换的矩阵

解: 设 $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ 满足 $AB = BA$, 即 $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

$$\Rightarrow \begin{bmatrix} a+c & b+d \\ c & d \end{bmatrix} = \begin{bmatrix} a & a+b \\ c & c+d \end{bmatrix} \Rightarrow \begin{cases} a+c=a \\ b+d=a+b \\ c=c+d \end{cases} \Rightarrow \begin{cases} c=0 \\ a=d \end{cases}$$

故所求矩阵 $B = \begin{bmatrix} a & b \\ 0 & a \end{bmatrix}$ (a, b 为任意常数)

5. 计算下列矩阵 (n 为正整数)

$$(1) \begin{bmatrix} 1 & 0 \\ \lambda & 1 \end{bmatrix}^n$$

$$\text{解: } A^2 = \begin{bmatrix} 1 & 0 \\ \lambda & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \lambda & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2\lambda & 1 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 1 & 0 \\ 2\lambda & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \lambda & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3\lambda & 1 \end{bmatrix}$$

$$\text{推测: } A^k = \begin{bmatrix} 1 & 0 \\ k\lambda & 1 \end{bmatrix}$$

当 $k=1$ 时显然成立

$$\text{假设 } k \text{ 时成立, 则 } A^{k+1} = A^k A = \begin{bmatrix} 1 & 0 \\ k\lambda & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \lambda & 1 \end{bmatrix} \\ = \begin{bmatrix} 1 & 0 \\ (k+1)\lambda & 1 \end{bmatrix}$$

$$\text{由数学归纳法: } A^n = \begin{bmatrix} 1 & 0 \\ n\lambda & 1 \end{bmatrix}$$

$$(2) \begin{bmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{bmatrix}^n$$

$$\text{解: } A^2 = \begin{bmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{bmatrix} \begin{bmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{bmatrix} = \begin{bmatrix} \lambda^2 & 2\lambda & 1 \\ 0 & \lambda^2 & 2\lambda \\ 0 & 0 & \lambda^2 \end{bmatrix}$$

$$A^3 = A^2 A = \begin{bmatrix} \lambda^3 & 3\lambda^2 & 3\lambda \\ 0 & \lambda^3 & 3\lambda^2 \\ 0 & 0 & \lambda^3 \end{bmatrix} \dots$$

$$\text{推测: } A^k = \begin{bmatrix} \lambda^k & k\lambda^{k-1} & \frac{k(k-1)}{2}\lambda^{k-2} \\ 0 & \lambda^k & k\lambda^{k-1} \\ 0 & 0 & \lambda^k \end{bmatrix}$$

当 $k=2$ 时显然成立

$$\text{假设 } k \text{ 时成立, 则 } A^{k+1} = A^k A = \begin{bmatrix} \lambda^{k+1} & (k+1)\lambda^k & \frac{(k+1)k}{2}\lambda^{k-1} \\ 0 & \lambda^{k+1} & (k+1)\lambda^k \\ 0 & 0 & \lambda^{k+1} \end{bmatrix}$$

$$\text{由数学归纳法: } A^n = \begin{bmatrix} \lambda^n & n\lambda^{n-1} & \frac{n(n-1)}{2}\lambda^{n-2} \\ 0 & \lambda^n & n\lambda^{n-1} \\ 0 & 0 & \lambda^n \end{bmatrix}$$

6. 设矩阵 $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{bmatrix}$, 求 $A^n - 2A^{n-1}$ ($n \geq 2$).

$$\text{解: } \because A^2 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 2 \\ 0 & 4 & 0 \\ 2 & 0 & 2 \end{bmatrix} = 2A$$

$$A^3 = A^2 \cdot A = 2A^2 \quad \dots \quad A^n = A^2 A^{n-2} = 2A \cdot A^{n-2} = 2A^{n-1}$$

$$\therefore A^n - 2A^{n-1} = 0$$

7. 已知 $\alpha = (1, 2, 3)$, $\beta = (1, \frac{1}{2}, \frac{1}{3})$, $A = \alpha^T \beta$, 求 A^n (n 为正整数)

$$\text{解: } A = \alpha^T \beta = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} (1, \frac{1}{2}, \frac{1}{3}) = \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ 2 & 1 & \frac{2}{3} \\ 3 & \frac{3}{2} & 1 \end{bmatrix} \quad \beta \alpha^T = (1, \frac{1}{2}, \frac{1}{3}) \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = 3$$

$$A^n = (\alpha^T \beta)^n = (\alpha^T \beta)(\alpha^T \beta) \dots (\alpha^T \beta) = \alpha^T (\underbrace{\beta \alpha^T \beta \alpha^T \dots \beta \alpha^T}_{(n-1) \uparrow}) \beta = 3^{n-1} \alpha^T \beta = 3^{n-1} A$$

$$= 3^{n-1} \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ 2 & 1 & \frac{2}{3} \\ 3 & \frac{3}{2} & 1 \end{bmatrix}$$

8. 设 A, B 都是 n 阶对称矩阵, 证明: AB 是对称矩阵的充分必要条件是 $AB = BA$

$$\text{证明: } \because A^T = A, \quad B^T = B$$

$$\text{若 } AB = BA, \text{ 则 } (AB)^T = B^T A^T = BA = AB \quad \therefore AB \text{ 是对称矩阵}$$

$$\text{若 } AB \text{ 是对称矩阵, 则 } (AB)^T = AB, \text{ 有 } AB = B^T A^T = BA$$

9. 设A是三阶矩阵, 若已知 $|A|=m$, 求 $|-mA|$

解: $|-mA| = (-m)^3 |A| = -m^4$

10. 设A为n阶矩阵, n为奇数, 且 $AA^T = E_n$, $|A|=1$, 求 $|A - E_n|$

解:

$$\begin{aligned} \therefore |A - E_n| &= |A - AA^T| = |A| |\underbrace{E_n - A^T}_{E_n - A}| = |A| |\underbrace{E_n - A}_{-(A - E_n)}| \\ &= (-1)^n |A - E_n| = -|A - E_n| \end{aligned}$$

$$\therefore |A - E_n| = 0$$

11. 设A, B均为n阶矩阵, 且 $A = \frac{1}{2}(B + E)$, 证明: $A^2 = A$ 当且仅当 $B^2 = E$

证明:

若 $A^2 = A$

则 $A^2 = \frac{1}{4}(B + E)^2$

$\therefore BE = EB = B$ $\therefore$ B与E可交换 故 $A^2 = \frac{1}{4}(B^2 + 2B + E) = \frac{1}{2}(B + E)$

$$B^2 + \cancel{2B} + E = \cancel{2B} + 2E \Rightarrow B^2 = E$$

若 $B^2 = E$

$$\text{则 } A^2 = \frac{1}{4}(B + E)^2 = \frac{1}{4}(B^2 + 2B + E) = \frac{1}{4}(E + 2B + E) = \frac{1}{2}(B + E) = A$$

逆矩阵

只有方阵才有伴随矩阵

对 n 阶方阵 A , 若存在 n 阶方阵 B , 使 $AB=BA=E$, 则称 A 是可逆矩阵, 并称矩阵 B 是 A 的逆矩阵, 记为 A^{-1} . $AB=E$ 和 $BA=E$ 其中之一成立, 则 A, B 均可逆且互为逆矩阵

若矩阵 A 可逆, 则 A 的逆矩阵是唯一的

证明: 设 B, C 均为 A 的逆矩阵, 即 $AB=BA=E, AC=CA=E$

$$\text{则 } B = BE = B(AC) = (BA)C = EC = C$$

故 A 的逆矩阵唯一

运算律

若 A 可逆, 则 A^{-1} 也可逆, 且 $(A^{-1})^{-1} = A$

若 A 可逆, 数 $\lambda \neq 0$, 则 λA 也可逆, 且 $(\lambda A)^{-1} = \lambda^{-1} A^{-1}$

若 A, B 为同阶可逆矩阵, 则 AB 也可逆, 且 $(AB)^{-1} = B^{-1} A^{-1}$

若 A 可逆, 则 A^T 也可逆, 且 $(A^T)^{-1} = (A^{-1})^T$

伴随矩阵

对方阵 A , 由其行列式 $|A|$ 各元素的代数余子式构成的方阵, 记作 A^*

$$\begin{bmatrix} A_{11} & A_{21} & \dots & A_{n1} \\ A_{12} & A_{22} & \dots & A_{n2} \\ \vdots & \vdots & \dots & \vdots \\ A_{1n} & A_{2n} & \dots & A_{nn} \end{bmatrix}$$

例 设 $A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ -3 & 2 & -5 \end{bmatrix}$

求 A 的伴随矩阵

解: $A_{11} = (-1)^{1+1} \begin{vmatrix} 1 & 0 \\ 2 & -5 \end{vmatrix} = -5$ $A_{12} = (-1)^{1+2} \begin{vmatrix} 2 & 0 \\ -3 & -5 \end{vmatrix} = 10$ $A_{13} = (-1)^{1+3} \begin{vmatrix} 2 & 1 \\ -3 & 2 \end{vmatrix} = 7$

$A_{21} = (-1)^{2+1} \begin{vmatrix} 0 & 1 \\ 2 & -5 \end{vmatrix} = 2$ $A_{22} = (-1)^{2+2} \begin{vmatrix} 1 & 1 \\ -3 & -5 \end{vmatrix} = -2$ $A_{23} = (-1)^{2+3} \begin{vmatrix} 1 & 0 \\ -3 & 2 \end{vmatrix} = -2$

$A_{31} = (-1)^{3+1} \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = -1$ $A_{32} = (-1)^{3+2} \begin{vmatrix} 1 & 1 \\ 2 & 0 \end{vmatrix} = 2$ $A_{33} = (-1)^{3+3} \begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix} = 1$

$$A^* = \begin{bmatrix} -5 & 2 & -1 \\ 10 & -2 & 2 \\ 7 & -2 & 1 \end{bmatrix}$$

$$AA^* = A^*A = |A|E$$

$|A|$ 是否为0与 A 是否可逆密切相关:

若方阵 A 的行列式 $|A|=0$, 则 A 为奇异阵, 否则, A 为非奇异阵

A 非奇异 ($|A| \neq 0$) 的充要条件: A 可逆且 $A^{-1} = \frac{1}{|A|} A^*$

例. 求二阶矩阵 $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ 的逆矩阵, 其中 $ad-bc \neq 0$

解: $\because |A| = ad-bc \neq 0 \quad \therefore A$ 可逆且 $A^{-1} = \frac{1}{|A|} A^*$

$$\text{又: } A^* = \begin{bmatrix} A_{11} & A_{21} \\ A_{12} & A_{22} \end{bmatrix} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \quad \therefore A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

例. 解矩阵方程 $\begin{bmatrix} 1 & 4 \\ -1 & 2 \end{bmatrix} X \begin{bmatrix} 2 & 0 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 0 & -1 \end{bmatrix}$

解: 记方程 $AXB=C$, 显然 A, B 均可逆

$$\begin{aligned} \therefore \overset{E}{A^{-1}} A X \overset{E}{B B^{-1}} &= A^{-1} C B^{-1}, \text{ 故 } X = A^{-1} C B^{-1} = \begin{bmatrix} 1 & 4 \\ -1 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 3 & 1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ -1 & 1 \end{bmatrix}^{-1} \\ &= \frac{1}{\begin{vmatrix} 1 & 4 \\ -1 & 2 \end{vmatrix}} \begin{bmatrix} 2 & -4 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 0 & -1 \end{bmatrix} \frac{1}{\begin{vmatrix} 2 & 0 \\ -1 & 1 \end{vmatrix}} \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ \frac{1}{4} & 0 \end{bmatrix} \end{aligned}$$

例. 设矩阵 $A = \begin{bmatrix} 4 & 2 & 3 \\ 1 & 1 & 0 \\ -1 & 2 & 3 \end{bmatrix}$ 满足 $AB = A + 2B$, 求矩阵 B .

还有
其他方法

$$\text{解: } \because AB = A + 2B \quad \therefore AB - 2B = A, \quad (A - 2E)B = A$$

$$\text{(初等变换)} \quad \text{又: } |A - 2E| = \begin{vmatrix} 4 & 2 & 3 \\ -1 & 1 & 0 \\ -1 & 2 & 3 \end{vmatrix} \xrightarrow{-2} \begin{vmatrix} 4 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = \begin{vmatrix} 2 & 2 & 3 \\ 1 & -1 & 0 \\ -1 & 2 & 1 \end{vmatrix} \xrightarrow{r_1 - 3r_3} \begin{vmatrix} 5 & -4 & 0 \\ 1 & -1 & 0 \\ -1 & 2 & 1 \end{vmatrix} = 1 \times (-1)^{3+2} \begin{vmatrix} 5 & -4 \\ 1 & -1 \end{vmatrix} = -1 \neq 0$$

$$\therefore A - 2E \text{ 可逆且 } (A - 2E)^{-1} = \frac{1}{|A - 2E|} (A - 2E)^*$$

$$= - \begin{bmatrix} -1 & 4 & 3 \\ -1 & 5 & 3 \\ 1 & -6 & -4 \end{bmatrix}$$

$$B = (A - 2E)^{-1} A = - \begin{bmatrix} -1 & 4 & 3 \\ -1 & 5 & 3 \\ 1 & -6 & -4 \end{bmatrix} \begin{bmatrix} 4 & 2 & 3 \\ 1 & 1 & 0 \\ -1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 3 & -8 & -6 \\ 2 & -9 & -6 \\ -2 & 12 & 9 \end{bmatrix}$$

$$A^2 = A \quad \not\Rightarrow \quad A = 0 \text{ 或 } A = E \quad (A \text{ 可逆时才成立})$$

$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ 满足 $A^2 = A$, 但 $A \neq 0, A \neq E$

$A^2 - A = 0 \quad A(A - E) = 0$ 若 A 可逆, 则 $A^{-1}A(A - E) = A^{-1} \cdot 0$ 即 $A - E = 0$, 才有 $A = E$

$$AX = AY \text{ 且 } A \neq 0 \quad \not\Rightarrow \quad X = Y \quad (A \text{ 可逆时才成立})$$

$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, y = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ 满足 $AX = AY$, 但 $x \neq y$

若 A 可逆, 则 $A^{-1}AX = A^{-1}AY$, 才有 $X = Y$

例. 设 $P = \begin{bmatrix} 1 & 2 \\ 1 & 4 \end{bmatrix}$, $\Lambda = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$, 并且 $AP = P\Lambda$, 求 A^n

解:

$$\because |P| = \begin{vmatrix} 1 & 2 \\ 1 & 4 \end{vmatrix} = 2 \neq 0 \quad \therefore P \text{ 可逆 且 } P^{-1} = \frac{1}{|P|} P^* = \frac{1}{2} \begin{bmatrix} 4 & -2 \\ -1 & 1 \end{bmatrix}$$

$$\because AP = P\Lambda \quad \therefore APP^{-1} = P\Lambda P^{-1}$$

$$\text{即 } A = P\Lambda P^{-1}, A^2 = P\Lambda P^{-1}P\Lambda P^{-1} = P\Lambda^2 P^{-1}, A^3 = P\Lambda^2 P^{-1}P\Lambda P^{-1} = P\Lambda^3 P^{-1} \dots$$

$$A^n = P\Lambda^n P^{-1}$$

$$\text{又: } \Lambda = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \Lambda^2 = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \times 2 \end{bmatrix}, \Lambda^3 = \begin{bmatrix} 1 & 0 \\ 0 & 2 \times 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 2^3 \end{bmatrix} \dots$$

$$\Lambda^n = \begin{bmatrix} 1 & 0 \\ 0 & 2^n \end{bmatrix}$$

$$\therefore A^n = \frac{1}{2} \begin{bmatrix} 1 & 2 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2^n \end{bmatrix} \begin{bmatrix} 4 & -2 \\ -1 & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 2^{n+1} \\ 1 & 2^{n+2} \end{bmatrix} \begin{bmatrix} 4 & -2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2-2^n & 2^n-1 \\ 2-2^{n+1} & 2^{n+1}-1 \end{bmatrix}$$

例设方阵 A 满足 $A^3 - 3A - 10E = 0$, 证明: A 和 $A - 4E$ 都可逆, 并求其逆矩阵

证明:

$$\text{由 } A^3 - 3A - 10E = 0 \quad \text{得} \quad A(A^2 - 3E) = 10E, \quad A \left(\frac{A^2 - 3E}{10} \right) = E$$

$$\text{即 } A \text{ 可逆 且 } A^{-1} = \frac{A^2 - 3E}{10}$$

$$\text{由 } A^3 - 3A - 10E = 0 \quad \text{得} \quad A^3 - 4A^2 + 4A^2 - 16A + 13A - 52E = -42E$$

$$(A - 4E)(A^2 + 4A + 13E) = -42E$$

$$(A - 4E) \left(-\frac{A^2 + 4A + 13E}{42} \right) = E$$

$$\text{即 } A - 4E \text{ 可逆 且 } (A - 4E)^{-1} = -\frac{A^2 + 4A + 13E}{42}$$

1. 求下列矩阵的逆矩阵

(1) $\begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$

解: $\begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}^{-1} = \frac{1}{\begin{vmatrix} 1 & 2 \\ 2 & 5 \end{vmatrix}} \begin{bmatrix} (-1)^{1+1} 5 & (-1)^{2+1} 2 \\ (-1)^{1+2} 2 & (-1)^{2+2} 1 \end{bmatrix}$
 $= \begin{bmatrix} 5 & -2 \\ -2 & 1 \end{bmatrix}$

(3) $\begin{bmatrix} 1 & 2 & -1 \\ 3 & 4 & -2 \\ 5 & -4 & 1 \end{bmatrix}$

解: $\begin{bmatrix} 1 & 2 & -1 \\ 3 & 4 & -2 \\ 5 & -4 & 1 \end{bmatrix}^{-1} = \frac{1}{\begin{vmatrix} 1 & 2 & -1 \\ 3 & 4 & -2 \\ 5 & -4 & 1 \end{vmatrix}} \begin{bmatrix} (-1)^{1+1}(-4)(-1)(-2)(-1)^{3+1}0 \\ (-1)^{1+2}3(-1)^{2+2}6(-1)^{3+2}1 \\ (-1)^{1+3}(-32)(-1)^{2+3}(-14)(-1)^{3+3}(-2) \end{bmatrix}$
 $= \frac{1}{\begin{vmatrix} 1 & 2 & -1 \\ 1 & 0 & 0 \\ 5 & -4 & 1 \end{vmatrix}} \begin{bmatrix} -4 & 2 & 0 \\ -13 & 6 & -1 \\ -32 & 14 & -2 \end{bmatrix}$
 $= \frac{1}{1 \times (-1)^{2+1} \begin{vmatrix} 2 & -1 \\ -4 & 1 \end{vmatrix}} \begin{bmatrix} -4 & 2 & 0 \\ -13 & 6 & -1 \\ -32 & 14 & -2 \end{bmatrix}$
 $= \begin{bmatrix} -2 & 1 & 0 \\ -\frac{13}{2} & 3 & -\frac{1}{2} \\ -16 & 7 & -1 \end{bmatrix}$

(2) $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

解: $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}^{-1} = \frac{1}{\begin{vmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{vmatrix}} \begin{bmatrix} \overset{④}{\cos\theta} & \overset{②}{\sin\theta} \\ \overset{③}{-\sin\theta} & \overset{①}{\cos\theta} \end{bmatrix}$
 $= \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$

(4) $\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \quad (a_1, a_2, \dots, a_n \neq 0)$

解: $\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}^{-1} = \frac{1}{\begin{vmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{vmatrix}} \begin{bmatrix} (-1)^{1+1}a_n & \dots & (-1)^{1+n}a_1 \\ \vdots & \ddots & \vdots \\ (-1)^{n+1}a_1 & \dots & (-1)^{n+n}a_n \end{bmatrix}$
 $= \begin{bmatrix} \frac{a_n}{a_1 a_2 \dots a_n} & \dots & \frac{a_1}{a_1 a_2 \dots a_n} \\ \vdots & \ddots & \vdots \\ \frac{a_1}{a_1 a_2 \dots a_n} & \dots & \frac{a_n}{a_1 a_2 \dots a_n} \end{bmatrix}$
 $= \begin{bmatrix} \frac{1}{a_1} & \frac{1}{a_2} & \dots & \frac{1}{a_n} \end{bmatrix}$

2. 利用逆矩阵解下列方程

(1) $\begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix} X = \begin{bmatrix} 4 & -6 \\ 2 & 1 \end{bmatrix}$

解: $\begin{vmatrix} 2 & 5 \\ 1 & 3 \end{vmatrix} \neq 0 \quad \therefore \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix} \text{ 可逆}$

$\begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}^{-1} \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix} X = \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}^{-1} \begin{bmatrix} 4 & -6 \\ 2 & 1 \end{bmatrix}$
 $= \frac{1}{\begin{vmatrix} 2 & 5 \\ 1 & 3 \end{vmatrix}} \begin{bmatrix} 3 & -5 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 4 & -6 \\ 2 & 1 \end{bmatrix}$
 $= \begin{bmatrix} 2 & -23 \\ 0 & 8 \end{bmatrix}$

(2) $X \begin{bmatrix} 2 & 1 & -1 \\ 2 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 3 \\ 4 & 3 & 2 \end{bmatrix}$

解: $\begin{vmatrix} 2 & 1 & -1 \\ 2 & 1 & 0 \\ 1 & -1 & 1 \end{vmatrix} \xrightarrow{r_2-r_1} \begin{vmatrix} 2 & 1 & -1 \\ 0 & 0 & 1 \\ 1 & -1 & 1 \end{vmatrix} = -\begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} \neq 0 \quad \therefore \begin{bmatrix} 2 & 1 & -1 \\ 2 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix} \text{ 可逆}$

$X \begin{bmatrix} 2 & 1 & -1 \\ 2 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & -1 \\ 2 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & -1 & 3 \\ 4 & 3 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 & -1 \\ 2 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix}^{-1}$

$X = \begin{bmatrix} 1 & -1 & 3 \\ 4 & 3 & 2 \end{bmatrix} \frac{1}{\begin{vmatrix} 2 & 1 & -1 \\ 2 & 1 & 0 \\ 1 & -1 & 1 \end{vmatrix}} \begin{bmatrix} (-1)^{1+1}1 & (-1)^{2+1}0 & (-1)^{3+1}1 \\ (-1)^{1+2}2 & (-1)^{2+2}3 & (-1)^{3+2}2 \\ (-1)^{1+3}(-3) & (-1)^{2+3}(-3) & (-1)^{3+3}0 \end{bmatrix}$
 $= \frac{1}{3} \begin{bmatrix} 1 & -1 & 3 \\ 4 & 3 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ -2 & 3 & -2 \\ -3 & 3 & 0 \end{bmatrix} = \begin{bmatrix} -2 & 2 & 1 \\ -\frac{8}{3} & 5 & -\frac{2}{3} \end{bmatrix}$

$$(3) \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -4 & 3 \\ 2 & 0 & -1 \\ 1 & -2 & 0 \end{bmatrix}$$

解: $\because \begin{vmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{vmatrix} = -1 \neq 0 \therefore \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ 可逆

$$\begin{aligned} X &= \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & -4 & 3 \\ 2 & 0 & -1 \\ 1 & -2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}^{-1} \\ &= \frac{1}{-1} \begin{bmatrix} 0 & (-1)^{2+1} & 1 & 0 \\ (-1)^{1+2} & 0 & 0 \\ 0 & 0 & (-1)^{3+3} \end{bmatrix} \begin{bmatrix} 1 & -4 & 3 \\ 2 & 0 & -1 \\ 1 & -2 & 0 \end{bmatrix} \begin{bmatrix} (-1)^{1+1} & 0 & 0 \\ 0 & 0 & (-1)^{2+2} \\ 0 & (-1)^{3+3} & 1 \end{bmatrix} \frac{1}{-1} \\ &= \begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & -4 & 3 \\ 2 & 0 & -1 \\ 1 & -2 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} -2 & 0 & 1 \\ -1 & 4 & -3 \\ -1 & 2 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 2 & -1 & 0 \\ 1 & 3 & -4 \\ 1 & 0 & -2 \end{bmatrix} \end{aligned}$$

3. 已知线性变换 $\begin{cases} x_1 = 2y_1 + 2y_2 + y_3 \\ x_2 = 3y_1 + y_2 + 5y_3 \\ x_3 = 3y_1 + 2y_2 + 3y_3 \end{cases}$, 求从变量 x_1, x_2, x_3 到变量 y_1, y_2, y_3 的线性变换

解: 系数矩阵 $\begin{bmatrix} 2 & 2 & 1 \\ 3 & 1 & 5 \\ 3 & 2 & 3 \end{bmatrix}$, $\begin{vmatrix} 2 & 2 & 1 \\ 3 & 1 & 5 \\ 3 & 2 & 3 \end{vmatrix} \xrightarrow[r_3-2r_2]{r_1-2r_2} \begin{vmatrix} -4 & 0 & -9 \\ 3 & 1 & 5 \\ -3 & 0 & -7 \end{vmatrix} = 1 \times (-1)^{2+2} \begin{vmatrix} -4 & -9 \\ -3 & -7 \end{vmatrix} = 28 - 27 = 1 \neq 0$

$\therefore \begin{bmatrix} 2 & 2 & 1 \\ 3 & 1 & 5 \\ 3 & 2 & 3 \end{bmatrix}$ 可逆

$$\begin{bmatrix} 2 & 2 & 1 \\ 3 & 1 & 5 \\ 3 & 2 & 3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 1 \\ 3 & 1 & 5 \\ 3 & 2 & 3 \end{bmatrix}^{-1} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \frac{1}{1} \begin{bmatrix} -7 & -4 & 9 \\ 6 & 3 & -7 \\ 3 & 2 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

即 $\begin{cases} y_1 = -7x_1 - 4x_2 + 9x_3 \\ y_2 = 6x_1 + 3x_2 - 7x_3 \\ y_3 = 3x_1 + 2x_2 - 4x_3 \end{cases}$

$$(4) \begin{cases} x_1 + 2x_2 + 3x_3 = 1 \\ 2x_1 + 2x_2 + 5x_3 = 2 \\ 3x_1 + 5x_2 + x_3 = 3 \end{cases}$$

对线性方程组 $Ax=b$ 来说, 若系数矩阵 A 可逆, 则 $x=A^{-1}b$. $\Leftarrow |A| \neq 0$

解: $\because$ 系数矩阵 $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 2 & 5 \\ 3 & 5 & 1 \end{bmatrix}$ 可逆

$$X = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 2 & 5 \\ 3 & 5 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$\begin{aligned} \therefore \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} &= \begin{bmatrix} 1 & 2 & 3 \\ 2 & 2 & 5 \\ 3 & 5 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \frac{1}{15} \begin{bmatrix} (-1)^{1+1}(-23) & (-1)^{2+1}(-12) & (-1)^{3+1}4 \\ (-1)^{1+2}4 & (-1)^{2+2}(-8) & (-1)^{3+2}(-1) \\ (-1)^{1+3}4 & (-1)^{2+3}(-1) & (-1)^{3+3}(-2) \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \\ &= \frac{1}{15} \begin{bmatrix} -23 & 12 & 4 \\ 4 & -8 & 1 \\ 4 & 1 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \text{ 故 } x_1=1, x_2=0, x_3=0 \end{aligned}$$

$$\text{或: } \begin{bmatrix} 1 & 2 & 3 & 1 \\ 2 & 2 & 5 & 2 \\ 3 & 5 & 1 & 3 \end{bmatrix} \xrightarrow[r_3-3r_1]{r_2-2r_1} \begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & -2 & -1 & 0 \\ 0 & -1 & -8 & 0 \end{bmatrix} \xrightarrow[r_2 \leftrightarrow r_3]{r_2-2r_3} \begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & -1 & -8 & 0 \\ 0 & 0 & 15 & 0 \end{bmatrix} \xrightarrow{r_3 \div 15} \begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & -1 & -8 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\xrightarrow[r_1-3r_3]{r_2+8r_3} \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \xrightarrow[r_2 \times (-1)]{r_1+2r_2} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

4. 若 $A^k = O$ (k 是正整数), 求证: $(E-A)^{-1} = E + A + A^2 + \dots + A^{k-1}$

证明:

$$\because (E-A)(E+A+A^2+\dots+A^{k-1}) = E + \cancel{A} + \cancel{A^2} + \dots + \cancel{A^{k-1}} - \cancel{A} - \cancel{A^2} - \dots - A^k = E - A^k = E$$

$$\therefore (E-A)^{-1} = E + A + A^2 + \dots + A^{k-1}$$

5. 设方阵 A 满足 $A^2 - A - 2E = O$, 证明: A 及 $A+2E$ 都可逆

证明:

$$\because A^2 - A - 2E = O \quad \therefore A^2 - A = 2E$$

$$\text{则 } |A^2 - A| = |2E| = 2$$

即 $|A||A-E| = 2$, 故 $|A| \neq 0$ $\therefore$ 矩阵 A 可逆

$$\text{而 } A+2E = A^2, \quad \therefore |A+2E| = |A^2| = |A|^2 \neq 0 \quad \therefore A+2E \text{ 可逆}$$

6. 若三阶矩阵 A 的伴随矩阵为 A^* , 已知 $|A| = \frac{1}{2}$, 求 $|(3A)^{-1} - 2A^*|$

解:

$$\because AA^* = A^*A = |A|E$$

$$\therefore A^* = |A|EA^{-1} = |A|A^{-1} = \frac{1}{2}A^{-1}$$

$$|(3A)^{-1} - 2A^*| = \left| \frac{1}{3}A^{-1} - 2A^* \right| = \left| \frac{1}{3}A^{-1} - A^{-1} \right| = \left| -\frac{2}{3}A^{-1} \right|$$

$$= \left(-\frac{2}{3} \right)^3 |A^{-1}| = \left(-\frac{2}{3} \right)^3 |A|^{-1} = -\frac{8}{27} \times 2 = -\frac{16}{27}$$

7. 若矩阵 A 可逆, 证明其伴随矩阵 A^* 也可逆, 且 $(A^*)^{-1} = (A^{-1})^*$

$$\because A^{-1} = \frac{1}{|A|}A^* \quad \therefore A^* = |A|A^{-1}$$

则 $|A^*| = |A||A^{-1}|$ $\because A$ 可逆, 有 A^{-1} 可逆 $\therefore |A| \neq 0, |A^{-1}| \neq 0, |A^*| \neq 0$

$$\therefore A^* \text{ 也可逆} \quad \text{且 } (A^*)^{-1} = (|A|A^{-1})^{-1} = \frac{1}{|A|}A$$

由伴随矩阵的性质 $AA^* = |A|E$, 有 $(A^{-1})(A^{-1})^* = |A^{-1}|E$

$$A(A^{-1})^* = |A^{-1}|E$$

$$\text{则 } (A^{-1})^* = A|A^{-1}| = A|A|^{-1} = \frac{1}{|A|}A$$

$$\text{故 } (A^*)^{-1} = (A^{-1})^*$$

8. 设 n 阶矩阵 A 的伴随矩阵 A^* , 证明:

(1) 若 $|A|=0$, 则 $|A^*|=0$

证明: ① 若 $A=0$, 则 $A^*=0$

由 $|A|=0$ 知 $|A^*|=0$

② 若 $A \neq 0$ 且 $|A|=0$,

假设 $|A^*| \neq 0$, 则有 $A^*(A^*)^{-1} = E$

由此得 $A = AA^*(A^*)^{-1} = |A|E(A^*)^{-1} = 0$

矛盾.

故 $|A^*|=0$.

(2) $|A^*| = |A|^{n-1}$

证明:

① 当 A 不可逆即 $|A|=0$ 时, $|A|^{n-1}$ 显然为 0

由 (1) 知 $|A^*|=0$.

故 $|A^*| = |A|^{n-1}$

② 当 A 可逆即 $|A| \neq 0$ 时

$A^{-1} = \frac{1}{|A|} A^*$ $A^* = A^{-1}|A|$

故 $|A^*| = |A|A^{-1}|A| = |A|^n|A|^{-1} = |A|^{n-1}$

9.

(1) 设 $A = \begin{bmatrix} 0 & 3 & 3 \\ 1 & 1 & 0 \\ -1 & 2 & 3 \end{bmatrix}$ 且 $AB = A + 2B$, 求 B

解: 由 $AB = A + 2B$ 得 $(A - 2E)B = A$

$\therefore A - 2E = \begin{bmatrix} -2 & 3 & 3 \\ 1 & -1 & 0 \\ -1 & 2 & 1 \end{bmatrix}$, $|A - 2E| = 2 \neq 0 \therefore A - 2E$ 是可逆矩阵

故 $(A - 2E)^{-1}(A - 2E)B = (A - 2E)^{-1}A$

$B = \frac{1}{|A - 2E|} \begin{bmatrix} -2 & 3 & 3 \\ 1 & -1 & 0 \\ -1 & 2 & 1 \end{bmatrix}^* \begin{bmatrix} 0 & 3 & 3 \\ 1 & 1 & 0 \\ -1 & 2 & 3 \end{bmatrix}$

$= \frac{1}{2} \begin{bmatrix} -1 & 3 & 3 \\ -1 & 1 & 3 \\ 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} 0 & 3 & 3 \\ 1 & 1 & 0 \\ -1 & 2 & 3 \end{bmatrix}$

$= \begin{bmatrix} 0 & 3 & 3 \\ -1 & 2 & 3 \\ 1 & 1 & 0 \end{bmatrix}$

10. 已知 $A^* = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & -3 & 0 & 8 \end{bmatrix}$, 并且 $ABA^{-1} = BA^{-1} + 3E$, 求 B .

解: 由 $ABA^{-1} = BA^{-1} + 3E$ 得 $(A - E)BA^{-1} = 3E \Rightarrow (A - E)B = 3A$

$\therefore |A^*| = (-1)^{t(1234)} a_{11}a_{22}a_{33}a_{44} = 1 \times 1 \times 1 \times 8 = 8 \neq 0 \therefore A^*$ 可逆

由 $|A^*| = |A|^{n-1}$ 得 $|A|^{4-1} = |A^*| = 8 \therefore |A| = 2$

~~$A^{-1} = \frac{1}{|A|} A^* = \frac{1}{2} A^*$~~

由 $AA^* = |A|E = 2E$ 得 $A = 2(A^*)^{-1} = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ -2 & 0 & 2 & 0 \\ 0 & \frac{3}{2} & 0 & \frac{1}{4} \end{bmatrix}$

故 $A - E = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -2 & 0 & 1 & 0 \\ 0 & \frac{3}{2} & 0 & -\frac{3}{4} \end{bmatrix}$, $|A - E| \neq 0$, $\therefore A - E$ 可逆 $(A - E)^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 2 & 0 & 1 & 0 \\ 0 & 1 & 0 & -\frac{4}{3} \end{bmatrix}$

又由 $(A - E)B = 3A$, 得 $B = 3(A - E)^{-1}A$

$= \begin{bmatrix} 6 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 \\ 6 & 0 & 6 & 0 \\ 0 & 3 & 0 & -1 \end{bmatrix}$

(2) 设 $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{bmatrix}$, 且 $AB + E = A^2 + B$, 求 B

解: 由 $AB + E = A^2 + B$ 整理可得 $(A - E)B = (A - E)(A + E)$

$\therefore A - E = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \neq 0$, $\therefore A - E$ 是可逆矩阵

故 $(A - E)^{-1}(A - E)B = (A - E)^{-1}(A - E)(A + E)$,

$B = A + E$

$= \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & 0 \\ 1 & 0 & 2 \end{bmatrix}$

或: $A^{-1}(A - E)B = 3A^{-1}A$

$(E - A^{-1})B = 3E$

$(E - \frac{1}{2}A^*)B = 3E$

$(2E - A^*)B = 6E$

$B = 6(2E - A^*)^{-1} = \dots$

11. 设 $P^{-1}AP = \Lambda$, 其中 $P = \begin{bmatrix} -1 & -4 \\ 1 & 1 \end{bmatrix}$, $\Lambda = \begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix}$, 求 A^n

解: $\because |P| = \begin{vmatrix} -1 & -4 \\ 1 & 1 \end{vmatrix} \neq 0 \therefore P$ 可逆, $P^{-1}AP = \Lambda \quad P^{-1}A^nP = P\Lambda^nP^{-1}$

$$\begin{aligned} A^n &= (P\Lambda P^{-1})^n \\ &= \underbrace{P\Lambda P^{-1} \cdot P\Lambda P^{-1} \cdots P\Lambda P^{-1}}_{n \uparrow} \\ &= P\Lambda^n P^{-1} \end{aligned}$$

$$\begin{aligned} A^n &= (P\Lambda P^{-1})^n = P\Lambda^n P^{-1} = \begin{bmatrix} -1 & -4 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix}^n \begin{bmatrix} -1 & -4 \\ 1 & 1 \end{bmatrix}^{-1} \\ &= \begin{bmatrix} -1 & -4 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 2^n \end{bmatrix} \frac{1}{\begin{vmatrix} -1 & -4 \\ 1 & 1 \end{vmatrix}} \begin{bmatrix} 1 & 4 \\ -1 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 2^{\frac{n+1}{2}} & 2^{\frac{n+1}{2}} \\ -683 & -684 \end{bmatrix} \end{aligned}$$

12. 设矩阵 A, B 及 $A+B$ 均可逆, 证明 $A^{-1} + B^{-1}$ 也可逆, 并求其逆矩阵.

证明: $\because A, B$ 以及 $A+B$ 均可逆 $\therefore AA^{-1} = A^{-1}A = E, BB^{-1} = B^{-1}B = E$
 $A^{-1} + B^{-1} = A^{-1}E + EB^{-1} = A^{-1}BB^{-1} + A^{-1}AB^{-1} = A^{-1}(A+B)B^{-1}$

$\therefore A^{-1} + B^{-1}$ 可表示为 3 个可逆矩阵的乘积 $\Rightarrow A^{-1} + B^{-1}$ 可逆

$$(A^{-1} + B^{-1})^{-1} = [A^{-1}(A+B)B^{-1}]^{-1} = (B^{-1})^{-1}(A+B)^{-1}(A^{-1})^{-1} = B(A+B)^{-1}A$$

13. 设 n 阶矩阵 A, B 满足 $A+B = AB$

(1) 证明: $A-E$ 为可逆矩阵

证明: 由 $A+B = AB$

$$\text{得 } A-E + (E-A)B = -E$$

$$\text{即 } (A-E)(B-E) = E$$

由此知 $A-E$ 可逆

(2) 已知 $B = \begin{bmatrix} 1 & -3 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$, 求 A

解: 由 $(A-E)(B-E) = E \quad \because |B-E| \neq 0 \therefore B-E$ 可逆

两边同乘 $(B-E)^{-1}$ 得 $A-E = (B-E)^{-1}$

$$\begin{aligned} A &= E + (B-E)^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & -3 & 0 \\ 2 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} \\ &= \begin{bmatrix} 1 & \frac{1}{2} & 0 \\ -\frac{1}{3} & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \end{aligned}$$

14. 设 A 为 n 阶非零矩阵, $A^* = A^T$, 其中 A^* 为 A 的伴随矩阵, 证明: A 可逆

$$\text{证明: 设 } A = (a_{ij})_{n \times n} \quad \because A^* = A^T, \quad A^T = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{n1} \\ a_{12} & a_{22} & \cdots & a_{n2} \\ a_{13} & a_{23} & \cdots & a_{n3} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{nn} \end{bmatrix} \quad A^* = \begin{bmatrix} A_{11} & A_{21} & \cdots & A_{n1} \\ A_{12} & A_{22} & \cdots & A_{n2} \\ A_{13} & A_{23} & \cdots & A_{n3} \\ \vdots & \vdots & \ddots & \vdots \\ A_{1n} & A_{2n} & \cdots & A_{nn} \end{bmatrix}$$

$$\therefore a_{ij} = A_{ij} \quad (i, j = 1, 2, \dots, n)$$

$\because A \neq 0$, 就设 $a_{11} \neq 0$ 于是 $|A| = a_{11}A_{11} + a_{12}A_{12} + \cdots + a_{1n}A_{1n}$

$$= a_{11}^2 + a_{12}^2 + \cdots + a_{1n}^2 > 0, \neq 0$$

故 A 可逆

分块矩阵

① 设矩阵A与B的行数、列数相同, 采用相同的分块方法

$$\text{若 } A = \begin{bmatrix} A_{11} & \dots & A_{1t} \\ \vdots & & \vdots \\ A_{s1} & \dots & A_{st} \end{bmatrix}, \quad B = \begin{bmatrix} B_{11} & \dots & B_{1t} \\ \vdots & & \vdots \\ B_{s1} & \dots & B_{st} \end{bmatrix}, \quad A+B = \begin{bmatrix} A_{11}+B_{11} & \dots & A_{1t}+B_{1t} \\ \vdots & & \vdots \\ A_{s1}+B_{s1} & \dots & A_{st}+B_{st} \end{bmatrix}$$

$$\textcircled{2} \quad kA = \begin{bmatrix} kA_{11} & \dots & kA_{1t} \\ \vdots & & \vdots \\ kA_{s1} & \dots & kA_{st} \end{bmatrix}$$

$$\textcircled{3} \quad \text{设 } A \text{ 为 } m \times l \text{ 矩阵, } B \text{ 为 } l \times n \text{ 矩阵, 分块成 } A = \begin{bmatrix} A_{11} & \dots & A_{1t} \\ \vdots & & \vdots \\ A_{s1} & \dots & A_{st} \end{bmatrix}, \quad B = \begin{bmatrix} B_{11} & \dots & B_{1r} \\ \vdots & & \vdots \\ B_{t1} & \dots & B_{tr} \end{bmatrix}$$

$$\text{则 } AB = \begin{bmatrix} C_{11} & \dots & C_{1r} \\ \vdots & & \vdots \\ C_{s1} & \dots & C_{sr} \end{bmatrix} \quad (\text{当 } A_{p1}, A_{p2}, \dots, A_{pt} \text{ 列数分别等于 } B_{1q}, B_{2q}, \dots, B_{tq} \text{ 行数时成立})$$

$$C_{pq} = \sum_{k=1}^t A_{pk} B_{kq} \quad (p=1, 2, \dots, s; q=1, 2, \dots, r)$$

例. 设矩阵 $A = \begin{bmatrix} 1 & 0 & 1 & 3 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 6 & 3 & 1 & 0 \\ 0 & -2 & 0 & 1 \end{bmatrix}$, 用分块矩阵计算 $kA, A+B$

解: 将矩阵A、B分块如下:

$$A = \begin{bmatrix} 1 & 0 & 1 & 3 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} E_2 & C \\ O_2 & -E_2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 6 & 3 & 1 & 0 \\ 0 & -2 & 0 & 1 \end{bmatrix} = \begin{bmatrix} D & O_2 \\ F & E_2 \end{bmatrix}$$

$$\text{则 } kA = k \begin{bmatrix} E_2 & C \\ O_2 & -E_2 \end{bmatrix} = \begin{bmatrix} kE_2 & kC \\ O_2 & -kE_2 \end{bmatrix} = \begin{bmatrix} k & 0 & k & 3k \\ 0 & k & 2k & 4k \\ 0 & 0 & -k & 0 \\ 0 & 0 & 0 & -k \end{bmatrix}$$

$$A+B = \begin{bmatrix} E_2 & C \\ O_2 & -E_2 \end{bmatrix} + \begin{bmatrix} D & O_2 \\ F & E_2 \end{bmatrix} = \begin{bmatrix} E_2+D & C \\ F & O_2 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 1 & 3 \\ 2 & 1 & 2 & 4 \\ 6 & 3 & 0 & 0 \\ 0 & -2 & 0 & 0 \end{bmatrix}$$

例. 设矩阵 $A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 2 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 & 1 & 0 \\ -1 & 2 & 0 & 1 \\ 1 & 0 & 4 & 1 \\ -1 & -1 & 2 & 0 \end{bmatrix}$, 用分块矩阵计算 AB .

解: 将A、B分块成 $A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 2 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} E & O \\ A_1 & E \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 & 1 & 0 \\ -1 & 2 & 0 & 1 \\ 1 & 0 & 4 & 1 \\ -1 & -1 & 2 & 0 \end{bmatrix} = \begin{bmatrix} B_{11} & E \\ B_{21} & B_{22} \end{bmatrix}$

$$AB = \begin{bmatrix} E & O \\ A_1 & E \end{bmatrix} \begin{bmatrix} B_{11} & E \\ B_{21} & B_{22} \end{bmatrix} = \begin{bmatrix} B_{11} & E \\ A_1 B_{11} + B_{21} & A_1 + B_{22} \end{bmatrix}$$

$$\text{而 } A_1 B_{11} + B_{21} = \begin{bmatrix} -1 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} -3 & 4 \\ 0 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} -2 & 4 \\ -1 & 1 \end{bmatrix}$$

$$A_1 + B_{22} = \begin{bmatrix} -1 & 2 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 4 & 1 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & 1 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & 0 & 1 & 0 \\ -1 & 2 & 0 & 1 \\ -2 & 4 & 3 & 3 \\ -1 & 1 & 3 & 1 \end{bmatrix}$$

设 $A = \begin{bmatrix} A_{11} & \cdots & A_{1r} \\ \vdots & & \vdots \\ A_{s1} & \cdots & A_{sr} \end{bmatrix}$, 则 $A^T = \begin{bmatrix} A_{11}^T & \cdots & A_{s1}^T \\ \vdots & & \vdots \\ A_{1r}^T & \cdots & A_{sr}^T \end{bmatrix}$

设 A 为 n 阶矩阵, 若 A 的分块矩阵只在对角线上有非零子块, 其余子块都为零矩阵, 且在对角线上的子块都是方阵, 则

$A = \begin{bmatrix} A_1 & & \\ & A_2 & \\ & & \ddots \\ & & & A_s \end{bmatrix}$, 其中 $A_i (i=1, 2, \dots, s)$ 都是方阵, 称 A 为分块对角矩阵

性质:

① 若 $|A_i| \neq 0 (i=1, 2, \dots, s)$, 则 $|A| \neq 0$ 且 $|A| = |A_1| |A_2| \cdots |A_s|$

② 若 $|A_i| \neq 0 (i=1, 2, \dots, s)$, 则

$$A^{-1} = \begin{bmatrix} A_1^{-1} & & \\ & A_2^{-1} & \\ & & \ddots \\ & & & A_s^{-1} \end{bmatrix}$$

③ 同结构的分块对角矩阵的和、差、积、数乘及逆仍是分块对角矩阵, 且运算表现为对应子块的运算

例 $A = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 1 \\ 0 & 2 & 1 \end{bmatrix}$, 求 A^{-1}

解: 将 A 分块如下: $A = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 1 \\ 0 & 2 & 1 \end{bmatrix} = \begin{bmatrix} A_{11} & 0 \\ 0 & A_{22} \end{bmatrix}$

则 $A_{11} = [5], A_{11}^{-1} = [\frac{1}{5}]$

$A_{22} = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} \quad |A_{22}| \neq 0 \quad \therefore A_{22} \text{ 可逆且 } A_{22}^{-1} = \frac{A_{22}^*}{|A_{22}|} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$

故 $A^{-1} = \begin{bmatrix} A_{11}^{-1} & 0 \\ 0 & A_{22}^{-1} \end{bmatrix} = \begin{bmatrix} \frac{1}{5} & 0 & 0 \\ 0 & 1 & -1 \\ 0 & -2 & 3 \end{bmatrix}$

按行分块与按列分块

$m \times n$ 矩阵 A 有 m 行, 称为矩阵 A 的 m 个行向量, 若第 i 行记作 $\alpha_i^T = (\alpha_{i1}, \alpha_{i2}, \dots, \alpha_{in})$

则 A 可记为 $A = \begin{bmatrix} \alpha_1^T \\ \alpha_2^T \\ \vdots \\ \alpha_m^T \end{bmatrix}$; 同理, 若第 j 列记作 $\beta_j = \begin{bmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{mj} \end{bmatrix}$, 则 $A = [\beta_1, \beta_2, \dots, \beta_n]$

对于 $A = (a_{ij})_{m \times s}$, $B = (b_{ij})_{s \times n}$ 的乘积矩阵 $AB = C = (c_{ij})_{m \times n}$,

若将 A 按行分为 m 块, 将 B 按列分为 n 块, 则

$$AB = \begin{bmatrix} \alpha_1^T \\ \alpha_2^T \\ \vdots \\ \alpha_m^T \end{bmatrix} [\beta_1, \beta_2, \dots, \beta_n] = \begin{bmatrix} \alpha_1^T \beta_1 & \alpha_1^T \beta_2 & \cdots & \alpha_1^T \beta_n \\ \alpha_2^T \beta_1 & \alpha_2^T \beta_2 & \cdots & \alpha_2^T \beta_n \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_m^T \beta_1 & \alpha_m^T \beta_2 & \cdots & \alpha_m^T \beta_n \end{bmatrix} = (c_{ij})_{m \times n},$$

其中 $c_{ij} = \alpha_i^T \beta_j = (\alpha_{i1}, \alpha_{i2}, \dots, \alpha_{is}) \begin{bmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{sj} \end{bmatrix} = \sum_{k=1}^s \alpha_{ik} b_{kj}$
 α 的行 β 的列

例. 设 $A^T A = 0$, 证明 $A = 0$

证明: 设 $A = (a_{ij})_{m \times n}$, 将 A 用列向量表示为 $A = [a_1, a_2, \dots, a_n]$

$$\text{则 } A^T A = \begin{bmatrix} a_1^T \\ a_2^T \\ \vdots \\ a_n^T \end{bmatrix} [a_1, a_2, \dots, a_n] = \begin{bmatrix} a_1^T a_1 & a_1^T a_2 & \dots & a_1^T a_n \\ a_2^T a_1 & a_2^T a_2 & \dots & a_2^T a_n \\ \vdots & \vdots & \ddots & \vdots \\ a_n^T a_1 & a_n^T a_2 & \dots & a_n^T a_n \end{bmatrix} = 0$$

则 $a_i^T a_j = 0$ ($i, j = 1, 2, \dots, n$), 从而 $a_j^T a_j = 0$ ($j = 1, 2, \dots, n$)

$$\text{又 } a_j^T a_j = [a_{1j} \ a_{2j} \ \dots \ a_{mj}] \begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix} = a_{1j}^2 + a_{2j}^2 + \dots + a_{mj}^2$$

$\therefore a_{1j}^2 + a_{2j}^2 + \dots + a_{mj}^2 = 0$ 故 $a_{ij} = 0$ ($i = 1, 2, \dots, m; j = 1, 2, \dots, n$).
即: $A = 0$.

克莱姆法则

将线性方程组

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \\ \dots \dots \dots \\ a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n = b_n \end{cases}, \text{记为 } AX = b.$$

其中 $A = (a_{ij})_{n \times n}$, $X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$, $b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$

若线性方程组 $AX = b$ 的系数行列式 $D = |A| \neq 0$, 则它有唯一解 $x_j = \frac{D_j}{D}$ ($j = 1, 2, \dots, n$)

证明: $\because |A| \neq 0 \therefore A^{-1}$ 存在, 从而 $X = A^{-1}b$

且由逆矩阵的唯一性知解唯一

又由 $A^{-1} = \frac{A^*}{|A|}$ 知 $X = A^{-1}b = \frac{A^*}{|A|}b$

即 $\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \frac{1}{D} \begin{bmatrix} a_{11} & a_{21} & \dots & a_{n1} \\ a_{12} & a_{22} & \dots & a_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$
 $= \frac{1}{D} \begin{bmatrix} a_{11}b_1 + a_{12}b_2 + \dots + a_{1n}b_n \\ a_{21}b_1 + a_{22}b_2 + \dots + a_{2n}b_n \\ \dots \dots \dots \\ a_{n1}b_1 + a_{n2}b_2 + \dots + a_{nn}b_n \end{bmatrix}$

如: $x_1 = \frac{D_1}{D}$ 代数余子式
 $= \frac{|a_{21} \ a_{22} \ \dots \ a_{2n}| \begin{vmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{vmatrix}}{|A|}$

$x_2 = \frac{D_2}{D}$
 $= \frac{|a_{12} \ a_{22} \ \dots \ a_{n2}| \begin{vmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{vmatrix}}{|A|}$
 $\dots$

$\therefore$ 亦即 $x_j = \frac{1}{D} (a_{1j}b_1 + a_{2j}b_2 + \dots + a_{nj}b_n) = \frac{D_j}{D}$ ($j = 1, 2, \dots, n$)

1. 用分块矩阵乘法计算 $\begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 3 & 0 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & -2 & 3 \\ 0 & 0 & 0 & -3 \end{bmatrix}$

解: 记 $A_{11} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$, $A_{22} = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$, $B_{12} = \begin{bmatrix} 3 & 0 \\ 2 & -1 \end{bmatrix}$, $B_{22} = \begin{bmatrix} -2 & 3 \\ 0 & -3 \end{bmatrix}$

则原式 = $\begin{bmatrix} A_{11} & E_2 \\ O_2 & A_{22} \end{bmatrix} \begin{bmatrix} E_2 & B_{12} \\ O_2 & B_{22} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{11}B_{12} + B_{22} \\ O_2 & A_{22}B_{22} \end{bmatrix}$

又 $A_{11}B_{12} + B_{22} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 2 & -1 \end{bmatrix} + \begin{bmatrix} -2 & 3 \\ 0 & -3 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} + \begin{bmatrix} -2 & 3 \\ 0 & -3 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 2 & -4 \end{bmatrix}$

$A_{22}B_{22} = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} -2 & 3 \\ 0 & -3 \end{bmatrix} = \begin{bmatrix} -4 & 3 \\ 0 & -9 \end{bmatrix}$

$\therefore A = \begin{bmatrix} 1 & 2 & -1 & 1 \\ 0 & 1 & 2 & -4 \\ 0 & 0 & -4 & 3 \\ 0 & 0 & 0 & -9 \end{bmatrix}$

2. 设 n 阶矩阵 A 及 s 阶矩阵 B 都可逆, 求:

(1) $\begin{bmatrix} O & A \\ B & O \end{bmatrix}^{-1}$

解: $\because A$ 与 B 均可逆

$\begin{bmatrix} O & A \\ B & O \end{bmatrix} \begin{bmatrix} O & B^{-1} \\ A^{-1} & O \end{bmatrix} = \begin{bmatrix} E_n & O \\ O & E_s \end{bmatrix} = E_{n+s}$

$\therefore \begin{bmatrix} O & A \\ B & O \end{bmatrix}$ 可逆.

且 $\begin{bmatrix} O & A \\ B & O \end{bmatrix}^{-1} = \begin{bmatrix} O & B^{-1} \\ A^{-1} & O \end{bmatrix}$

(3) $\begin{bmatrix} A & C \\ O & B \end{bmatrix}^{-1}$

$\downarrow$

同理:

$\begin{bmatrix} A^{-1} & -A^{-1}CB^{-1} \\ O & B^{-1} \end{bmatrix}$

(2) $\begin{bmatrix} A & O \\ C & B \end{bmatrix}^{-1}$ ★ 分块一致

解: 令 $\begin{bmatrix} A & O \\ C & B \end{bmatrix} \begin{bmatrix} X_{11} & X_{12} \\ X_{21} & X_{22} \end{bmatrix} = E_{n+s}$

其中, X_{11} 是 $n \times n$ 矩阵, X_{22} 是 $s \times s$ 矩阵, X_{12} 是 $n \times s$ 矩阵, X_{21} 是 $s \times n$ 矩阵

$\therefore \begin{bmatrix} A & O \\ C & B \end{bmatrix} \begin{bmatrix} X_{11} & X_{12} \\ X_{21} & X_{22} \end{bmatrix} = \begin{bmatrix} AX_{11} & AX_{12} \\ CX_{11} + BX_{21} & CX_{12} + BX_{22} \end{bmatrix} = \begin{bmatrix} E_n & O \\ O & E_s \end{bmatrix}$

由 $AX_{11} = E_n$, 得 $X_{11} = A^{-1}$

由 $AX_{12} = O$, 得 $X_{12} = O$

由 $CX_{11} + BX_{21} = O$, 得 $BX_{21} = -CX_{11} = -CA^{-1} \Rightarrow X_{21} = -B^{-1}CA^{-1}$

由 $CX_{12} + BX_{22} = E_s$, 得 $BX_{22} = E_s \Rightarrow X_{22} = B^{-1}$

$\therefore \begin{bmatrix} A & O \\ C & B \end{bmatrix}^{-1} = \begin{bmatrix} A^{-1} & O \\ -B^{-1}CA^{-1} & B^{-1} \end{bmatrix}$

3. 用矩阵的分块求下列矩阵的逆矩阵

$$(1) \begin{bmatrix} 5 & 2 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & 0 & 8 & 3 \\ 0 & 0 & 5 & 2 \end{bmatrix}$$

解: 记 $A_{11} = \begin{bmatrix} 5 & 2 \\ 2 & 1 \end{bmatrix}$, $A_{22} = \begin{bmatrix} 8 & 3 \\ 5 & 2 \end{bmatrix}$

则 $A = \begin{bmatrix} A_{11} & O_2 \\ O_2 & A_{22} \end{bmatrix}$ 且 $|A_{11}| \neq 0$, $|A_{22}| \neq 0$

故 $A^{-1} = \begin{bmatrix} A_{11}^{-1} & \\ & A_{22}^{-1} \end{bmatrix}$ 而 $A_{11}^{-1} = \frac{1}{\begin{vmatrix} 5 & 2 \\ 2 & 1 \end{vmatrix}} \begin{bmatrix} 1 & -2 \\ -2 & 5 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ -2 & 5 \end{bmatrix}$

$$A_{22}^{-1} = \frac{1}{\begin{vmatrix} 8 & 3 \\ 5 & 2 \end{vmatrix}} \begin{bmatrix} 2 & -3 \\ -5 & 8 \end{bmatrix} = \begin{bmatrix} 2 & -3 \\ -5 & 8 \end{bmatrix}$$

$$\therefore A^{-1} = \begin{bmatrix} 1 & -2 & 0 & 0 \\ -2 & 5 & 0 & 0 \\ 0 & 0 & 2 & -3 \\ 0 & 0 & -5 & 8 \end{bmatrix}$$

4. 设 A 为 3×3 矩阵, $|A| = -2$, 把 A 按列分块为 $A = [A_1 \ A_2 \ A_3]$, 其中 A_j ($j=1, 2, 3$) 为 A 的第 j 列, 求:

(1) $|A_1, 2A_2, A_3|$

解:

$$|A_1, 2A_2, A_3|$$

$$= 2 |A_1, A_2, A_3|$$

$$= 2 |A|$$

$$= 2 \times (-2)$$

$$= -4$$

(2) $|A_3 - 2A_1, 3A_2, A_1|$

解:

$$|A_3 - 2A_1, 3A_2, A_1|$$

$$= |A_3, 3A_2, A_1|$$

$$= 3 |A_3, A_2, A_1|$$

$$= -3 |A_1, A_2, A_3|$$

$$= -3 |A|$$

$$= -3 \times (-2) = 6$$

5. 设 A 为 n 阶矩阵, $\beta_1, \beta_2, \dots, \beta_n$ 为 A 的列子块, 试用 $\beta_1, \beta_2, \dots, \beta_n$ 表示 $A^T A$

解: $A^T A = [\beta_1 \ \beta_2 \ \dots \ \beta_n]^T [\beta_1 \ \beta_2 \ \dots \ \beta_n]$

$$= \begin{bmatrix} \beta_1^T \\ \beta_2^T \\ \vdots \\ \beta_n^T \end{bmatrix} [\beta_1 \ \beta_2 \ \dots \ \beta_n]$$

$$= \begin{bmatrix} \beta_1^T \beta_1 & \beta_1^T \beta_2 & \dots & \beta_1^T \beta_n \\ \beta_2^T \beta_1 & \beta_2^T \beta_2 & \dots & \beta_2^T \beta_n \\ \vdots & \vdots & \ddots & \vdots \\ \beta_n^T \beta_1 & \beta_n^T \beta_2 & \dots & \beta_n^T \beta_n \end{bmatrix}$$

$$(2) \begin{bmatrix} 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 2 & 8 \\ 1 & 0 & 1 & 0 & 0 \\ 2 & 3 & 2 & 0 & 0 \\ 3 & 1 & 1 & 0 & 0 \end{bmatrix}$$

解: 记 $A_{12} = \begin{bmatrix} 1 & 3 \\ 2 & 8 \end{bmatrix}$, $A_{21} = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 3 & 2 \end{bmatrix}$

则 $A = \begin{bmatrix} O & A_{12} \\ A_{21} & O \end{bmatrix}$ 且 $|A_{12}| \neq 0$, $|A_{21}| \neq 0$

故 $A^{-1} = \begin{bmatrix} O & A_{12}^{-1} \\ A_{21}^{-1} & O \end{bmatrix}$ 而 $A_{12}^{-1} = \begin{bmatrix} 4 & -\frac{3}{2} \\ -1 & \frac{1}{2} \end{bmatrix}$

$$A_{21}^{-1} = \begin{bmatrix} -\frac{1}{6} & -\frac{1}{6} & \frac{1}{2} \\ -\frac{2}{3} & \frac{1}{3} & 0 \\ \frac{1}{6} & \frac{1}{6} & -\frac{1}{2} \end{bmatrix}$$

$$\therefore A^{-1} = \begin{bmatrix} 0 & 0 & -\frac{1}{6} & -\frac{1}{6} & \frac{1}{2} \\ 0 & 0 & -\frac{2}{3} & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{6} & \frac{1}{6} & -\frac{1}{2} \\ 4 & -\frac{3}{2} & 0 & 0 & 0 \\ -1 & \frac{1}{2} & 0 & 0 & 0 \end{bmatrix}$$